Kron's field network and his 1962 polyhedron

Kron's equivalent circuit of Maxwell's equations and his curve-fitting polyhedron, checked against every number he printed. The standard-physics half of register entry H20 (step 1)

kronspice simulation report · 2026-11-07

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Summary

Register entry H20 reads Kron's polyhedral waves (1962) as a description of a physical field region. Its first step is to build the structures and check them against standard physics and against Kron's own numbers. This example solves Kron's 1945 field network (a cavity and a waveguide lattice) and the curve-fitting polyhedron of his 1962 paper with the circuit engine, and compares the results with every number he printed (figures and tables transcribed in *doc/kron-figure-catalogue.md*).

- **The 1945 cavity matches his network analyzer** to a worst 5.9 % of the largest potential (mean 2 %); his own unbalance was 8 %.
- **The TE_{11} box resonance** of the lattice equals the spatial part of Yee's dispersion to 2×10^{-14} and the continuum to 0.36 % (second order); Kron's solve equals MNA to 4×10^{-10}.
- **Kron 1962 Tables I–VI are reproduced:** given columns to 1.9×10^{-6}, the first planar frame to 8.9×10^{-6}, the oscillatory column to 2.8×10^{-5}.
- **A planar reference frame is a vector of whole numbers.** The genetic algorithm recovers its nine integers from the printed column alone (17,702 circuit evaluations).
- **The oscillatory column returns its input.** Its frame fits any data, 1000 random sets to 7×10^{-15}, so its agreement shows no estimating power and does not by itself support the physical reading of H20.
- **Power input is the error-square** (Kron p. 67): a one-mesh circuit's dissipation equals the summed error-square, ledger residual 4×10^{-13} W.

What is being tested

The hypothesis (H20). In his 1962 curve-fitting paper Kron built a k-dimensional polyhedron whose q-networks are copies of his 1944 Maxwell network, even-dimensional ones magnetic and odd-dimensional ones dielectric, joined to a dual polyhedron. He wrote that a plasma-filled field region "crystallizes" into such a structure, that it can oscillate self-adaptively, and that crystals realize it. Kron himself called the physical interpretation "only tentative" (p. 66); the hypothesis is the physical reading the user placed in the register.

Standard prediction. In three dimensions the polyhedron is a discretization of Maxwell's equations (Kron 1944; Tonti's cell method) and agrees with FDTD and TLM to discretization error; a passive linear structure has fixed modes; X-ray-excited crystal waves follow Ewald–Laue dynamical diffraction. This example covers the standard half of step 1: does the network reproduce Kron's printed numbers and standard field results?

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What was checked against what

source	kronspice	result
Kron 1945 Fig. 14, cavity potentials read on the network analyzer	the lattice solved exactly (kron.Kron1945Cavity)	worst 5.9 % of the largest potential, mean 2 %; his own unbalance was 8 %
Kron 1945 Fig. 15, unbalanced currents	recomputed from his Fig. 14 by his eq. (1)	23 of 25 junctions within 0.33 A; the other two look like his slips (below)
Balasubramanian 1968 Fig. 15, waveguide matching error	the TE lattice (kron.Waveguide), with a closed-form match leaving ripple 1e-16	+0.69, +1.50, +3.47, +6.43 % against about 1.0, 1.6, 3.4, 6.2 % read from his curve
TE_{11} box resonance	lattice dispersion, the spatial part of Yee's	equal to 2e-14; continuum to 0.36 %, second order; Kron's solve equals MNA to 4e-10
Kron 1962 Tables I–VI, given columns	CurveFit.Characteristics on his printed z	worst 1.9e-6
Tables I–VI, first planar frame	the integer frame below	worst 8.9e-6
Tables I–VI, oscillatory column	the closed-form frame below	worst 2.8e-5
Kron 1962 p. 67, power input is the error-square	one-mesh circuit, MNA ledger and Kron's mesh form	dissipation equals the summed error-square; ledger residual 4e-13 W

The two findings

A planar reference frame is a vector of whole numbers

Kron never states his frames. With 13 rows and 12 coefficients, a fit has one residual direction w, and $\hat{y} = y - w(w^T y)/(w^T w)$ whichever 12 columns span the model (y is the data column, $\hat{y}$ the estimate). The errors in the first planar column have rank one across the six tables, and that direction is

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w = ( -3, 1, 1, 1 |  4, -1, -3, -4, -4, 2 over the branch lengths |  2, -10, -4 over 9 )
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The genetic algorithm then recovers these nine integers from the printed column alone, with every candidate solved as a circuit (17,702 evaluations).

The oscillatory column returns its input

The oscillatory column equals the given column for all six functions, to rounding. All 13 characteristics are linear in the four values at the points, so they span only a 4-dimensional subspace. A frame with w orthogonal to that subspace fits every function exactly, with zero power, and *kron.CurveFit.OscillatoryFrame* writes one down:

$$N = (9,\ M_2/2,\ M_1/2,\ -13,\ -10,\ M_3/2),\ \text{any even } M$$

- That is why the oscillatory polyhedron "can estimate satisfactorily even a set of random numbers", of which Kron wrote "the significance of this is not yet understood" (p. 69): 1000 random sets here fit to 7×10^{-15}. It is also why six "widely different" functions fit without adjustment.
- A model with 12 coefficients is fitting 4 independent numbers, so this agreement shows no estimating power. It is a property of the frame, and it does not by itself support the physical reading of H20.
- This is an inference from the tables, with three limits: any member of the family matches the printed column, so which frame Kron used cannot be determined; he wrote that "no exact fit is possible", and a residual below about 10^{-5} cannot be excluded from the printed figures; Kron 1959 and the 1960 General Electric report have not been read and might state otherwise.

Where the GA did not succeed

On Kron's objective, minimum power input, the search over integer frames did not reach zero. In a sweep of 200 × 300 runs, none of 17 completed runs (three mutation steps, up to six seeds each) got there. The best was 3.7×10^{-4} W, against 90 W for frame 1. The sweep was cut off by its time limit before the 0.2-decade step ran. Zero power is a linear condition, so it is solved by the closed form, and the GA adds nothing there. *TestGAMinimumPower* records the shortfall.

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What is not reproduced

- **The second planar frame.** Its function rows follow frame 1's residual times a constant 0.3032, but its divided differences are not one linear map of z. That means several misprints or a different frame per table; six tables cannot separate the two.
- **Least-square + linear.** With two corrections it is linear to 2×10^{-5}. Two outer branches come out with the sign reversed and the third exact, but the inner rows have a rank-3 error map, not a single residual direction, so no frame of this form gives it.
- **The full 3-D cell with its ideal transformers** (Kron 1962 Fig. 1) is built in *kron/maxwell3d.go* and its tests, not in this example. Still open from stage B of the catalogue's plan: the TLM twin.
- **q-networks in general.** *kron.CurveFit* handles this one polyhedron. The second-difference formula S = −(Δz/L)(A/L) is read off the tables, not derived from Kron's construction.

Apparent errors in the printed sources

These are inferences from internal consistency, not published errata.

where	printed	needed / computed	note
Table II, least-square + linear, second function row	−12.339281	−12.330283	
Table I, same column, fourth divided difference	−6.4341227	−0.64341227	a slipped decimal point
1945 Fig. 15, row 2 column 3	2.15	−2.16	the figure's own e_u is −0.51 (a dropped minus sign)
1945 Fig. 15, row 1 column 0	2.12	−5.94 from the printed potentials	cause unresolved

Kron's p. 62 claim that a polyhedron "improves always" the least-square estimate fails in his own Table I: z_1 is linear, the least-square fit is exact, and frame 1 moves it from 9.000 to 8.879.

Status

H20 remains **untested** in the register: the physical claims (plasma, crystals, self-adaptive oscillation) have no equations in any source and no test yet. The standard half of step 1 is partly done: the 2-D lattice and the first planar frame reproduce Kron's numbers, the 3-D cell (in *kron/maxwell3d.go*) equals Yee's equations, and the TLM twin, the second planar frame and the least-square + linear column are still open. The oscillatory column's fit, which matches any values including random numbers, is not evidence for the physical reading.

Reproducing the results

```
go run ./examples/kron_polyhedron
go test ./examples/kron_polyhedron ./kron
```

Sources: G. Kron, J. Math. Anal. Appl. 5, 46–69 (1962); G. Kron, J. Appl. Phys. 16, 172–186 (1945); N. V. Balasubramanian, PhD thesis, Liverpool, 1968; H. Nicholson, ACSE Research Report 26, Sheffield, 1974; E. Tonti, PIER 32, 1–44 (2001). The example has no run_output.txt or figures; numbers are those recorded in its README.

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